

Properties of Logarithms

Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Apply the addition, subtraction, and power properties of logarithms to expand expressions
- Condense multiple logarithmic terms into a single logarithm using log properties
- Combine all three properties to simplify complex logarithmic expressions

Problems

1. Use the addition property to expand the following logarithm.

$$\log_5 (3x)$$

2. Use the subtraction property to expand the following logarithm.

$$\log_4 \left(\frac{x}{9} \right)$$

3. Use the power property to expand the following logarithm.

$$\log_3 (x^5)$$

4. Use the addition and power properties to fully expand the following logarithm.

$$\log_6 (4x^3)$$

5. Condense the following expression into a single logarithm using the addition property.

$$\log_7 (x) + \log_7 (y)$$

6. Condense the following expression into a single logarithm using the subtraction property.

$$\log_{11} (x) - \log_{11} (5)$$



7. Condense the following expression into a single logarithm.

$$3\log_2(x) + \log_2(5)$$

8. Fully expand the following logarithm using all necessary properties. The base is 10 (common log).

$$\log\left(\frac{x^2y}{z^3}\right)$$

9. Condense the following into a single logarithm.

$$2\log_8(x) - 4\log_8(y) + \log_8(3)$$

10. Fully expand the following complex logarithm.

$$\log_8\left(\frac{4(x-5)^2}{x^4(x-3)^3}\right)$$



Properties of Logarithms — Answer Key

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Answer Key

1. Answer: $\log_5(3) + \log_5(x)$

- Apply the addition property: $\log_B(M \cdot N) = \log_B(M) + \log_B(N)$
- Split $3x$ into two factors: $\log_5(3) + \log_5(x)$

2. Answer: $\log_4(x) - \log_4(9)$

- Apply the subtraction property: $\log_B(M/N) = \log_B(M) - \log_B(N)$
- Result: $\log_4(x) - \log_4(9)$

3. Answer: $5 \log_3(x)$

- Apply the power property: $\log_B(M^n) = n \cdot \log_B(M)$
- Bring the exponent 5 in front: $5 \cdot \log_3(x)$

4. Answer: $\log_6(4) + 3 \log_6(x)$

- Apply the addition property: $\log_6(4) + \log_6(x^3)$
- Apply the power property to $\log_6(x^3)$: $\log_6(4) + 3 \cdot \log_6(x)$

5. Answer: $\log_7(xy)$

- Apply the addition property in reverse: $\log_B(M) + \log_B(N) = \log_B(M \cdot N)$
- Combine: $\log_7(x \cdot y) = \log_7(xy)$

6. Answer: $\log_{11}(x/5)$

- Apply the subtraction property in reverse: $\log_B(M) - \log_B(N) = \log_B(M/N)$
- Combine: $\log_{11}(x/5)$

7. Answer: $\log_2(5x^3)$

- Apply the power property in reverse to $3 \cdot \log_2(x)$: $\log_2(x^3)$
- Apply the addition property: $\log_2(x^3) + \log_2(5) = \log_2(5x^3)$

8. Answer: $2 \log(x) + \log(y) - 3 \log(z)$

- Apply the subtraction property: $\log(x^2 \cdot y) - \log(z^3)$
- Apply the addition property to the numerator: $\log(x^2) + \log(y) - \log(z^3)$
- Apply the power property: $2 \cdot \log(x) + \log(y) - 3 \cdot \log(z)$

9. Answer: $\log_8(3x^2 / y^4)$

- Apply the power property: $\log_8(x^2) - \log_8(y^4) + \log_8(3)$
- Apply the addition property: $\log_8(3x^2) - \log_8(y^4)$
- Apply the subtraction property: $\log_8(3x^2 / y^4)$

10. Answer: $\log_8(4) + 2 \cdot \log_8(x-5) - 4 \cdot \log_8(x) - 3 \cdot \log_8(x-3)$





- Expand the numerator using the addition property: $\log_8(4) + \log_8[(x-5)^2]$
 - Expand the denominator using the addition property: $\log_8(x^4) + \log_8[(x-3)^3]$
 - Apply the power property throughout: $\log_8(4) + 2 \cdot \log_8(x-5) - [4 \cdot \log_8(x) + 3 \cdot \log_8(x-3)]$
 - Distribute the negative sign: $\log_8(4) + 2 \cdot \log_8(x-5) - 4 \cdot \log_8(x) - 3 \cdot \log_8(x-3)$
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